

AUDIT AND CONTROL-ASSURANCE REPORT

```

icdfa@icdfa: ~
File Actions Edit View Help

icdfa@icdfa:~$ sudo systemctl restart auditd
icdfa@icdfa:~$ sudo auditctl -l
-rwx /etc/passwd -p rwx -k passwd_changes
-rwx /etc/shadow -p rwx -k shadow_changes
-a always,exit -F archb64 -S execve -F key=program_execution
-a always,exit -F archb32 -S execve -F key=program_execution
-w /var/log/auth.log -p wa -k auth_failures

icdfa@icdfa:~$

```

Activity 1.3 Generate and Query Audit Events

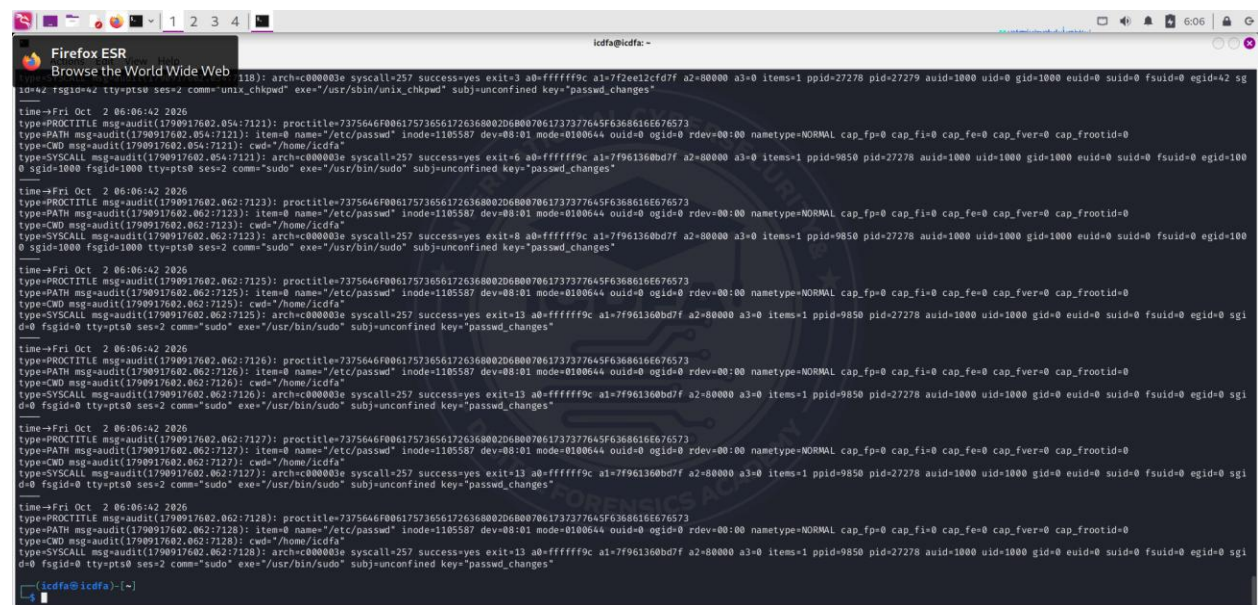
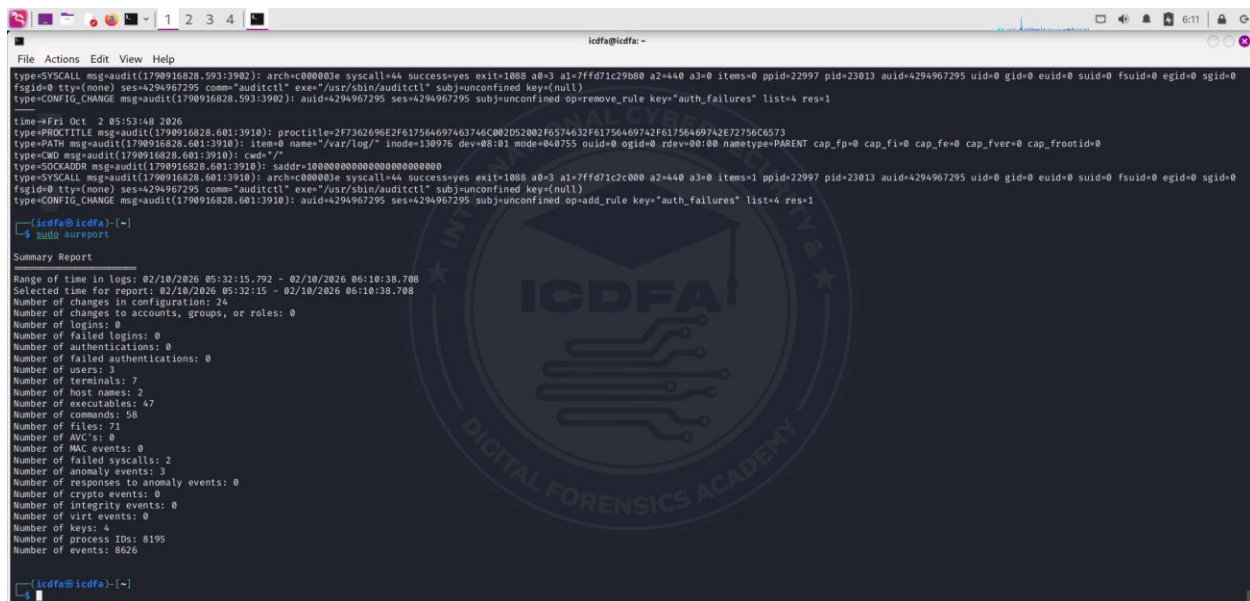


Figure 1-3_ausearch_password



```
File Actions Edit View Help
icdfa@icdfa:~$
type=SYSCALL msg=audit(1790916828.593:3902): arch=c000003e syscall=44 success=yes exit=1088 a0=3 a1=7ffd71c29b00 a2=440 a3=0 items=0 ppid=22997 pid=23013 auid=4294967295 uid=0 gid=0 euid=0 suid=0 fsuid=0 egid=0 sgid=0
fsuid=0 tty=(none) ses=4294967295 comm="auditctl" exe="/usr/sbin/auditctl" subj=unconfined key=(null)
type=CONFIG_CHANGE msg=audit(1790916828.593:3902): auid=4294967295 ses=4294967295 subj=unconfined op=remove_rule key="auth_failures" list=4 res=1

time=>Fri Oct 2 05:53:48 2026
type=PROCTITLE msg=audit(1790916828.601:3910): proctitle=2F73626962F617564697463746C002052002F6574632F61756469742E72756C6573
type=PATH msg=audit(1790916828.601:3910): item=0 name="/var/log/" inode=130976 dev=08:01 mode=040755 ouid=0 ogid=0 rdev=00:00 nametype=PARENT cap_fp=0 cap_fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(1790916828.601:3910): cwd="/"
type=SOCKADDR msg=audit(1790916828.601:3910): saddr=10000000000000000000000000000000
type=SYSCALL msg=audit(1790916828.601:3910): arch=c000003e syscall=44 success=yes exit=1088 a0=3 a1=7ffd71c2c000 a2=440 a3=0 items=1 ppid=22997 pid=23013 auid=4294967295 uid=0 gid=0 euid=0 suid=0 fsuid=0 egid=0 sgid=0
fsuid=0 tty=(none) ses=4294967295 comm="auditctl" exe="/usr/sbin/auditctl" subj=unconfined key=(null)
type=CONFIG_CHANGE msg=audit(1790916828.601:3910): auid=4294967295 ses=4294967295 subj=unconfined op=add_rule key="auth_failures" list=4 res=1

~(icdfa@icdfa):~$
~(icdfa@icdfa):~$ sudo aureport
Summary Report
Range of time in logs: 02/10/2026 05:32:15.792 - 02/10/2026 06:10:38.700
Selected time for report: 02/10/2026 05:32:15 - 02/10/2026 06:10:38.708
Number of changes in configuration: 24
Number of changes to accounts, groups, or roles: 0
Number of logins: 0
Number of failed logins: 0
Number of authentications: 0
Number of failed authentications: 0
Number of users: 3
Number of terminals: 7
Number of host names: 2
Number of executables: 47
Number of commands: 58
Number of files: 71
Number of AVC's: 0
Number of MAC events: 0
Number of failed syscalls: 2
Number of anomaly events: 3
Number of responses to anomaly events: 0
Number of crypto events: 0
Number of integrity events: 0
Number of virt events: 0
Number of keys: 4
Number of process IDs: 8195
Number of events: 8526

~(icdfa@icdfa):~$
```

Figure 1-3_aureport_summary

A short Interpretation explaining what the observed audit record says

At 06:06:42 on 2 October 2026, auditd recorded that the `/etc/passwd` file was accessed under the key `passwd_changes`.

The user was `icdfa` with `auid=1000`, and the command was run as root (`uid=0`) using `sudo` from terminal `pts0`. The action was `openat` and it was successful.

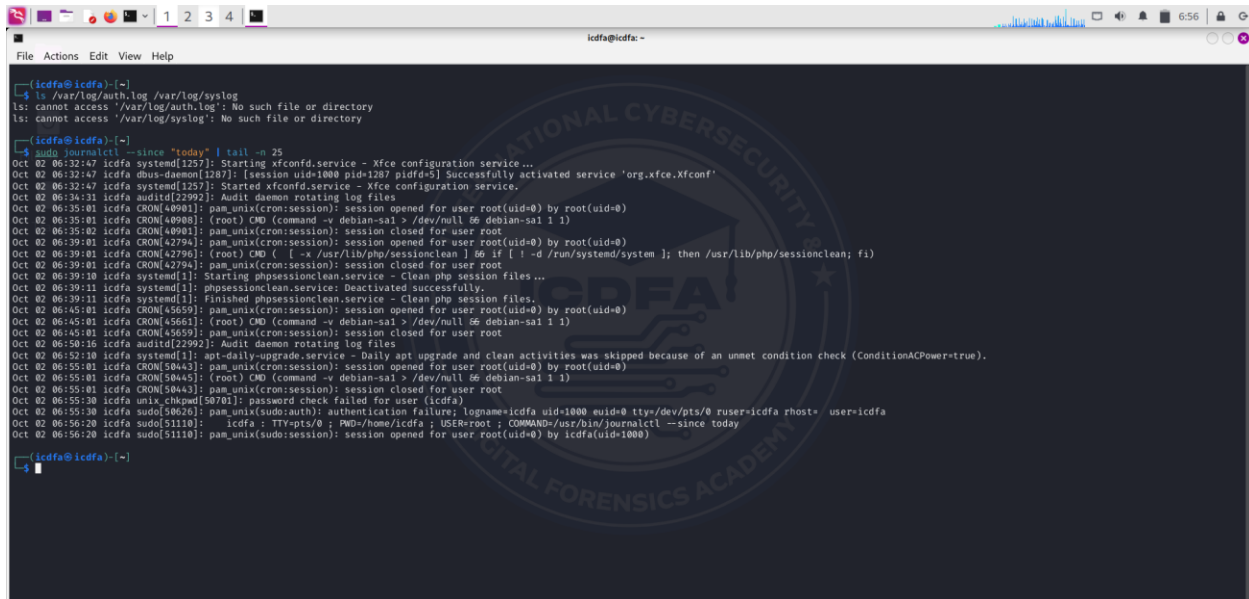
Even though the command was running with root privileges, the audit log still kept the original user ID, which means we can trace the activity back to the actual user who performed it.

The file was only opened for reading, so no changes were made to `/etc/passwd`. This is important from a governance and accountability perspective because `/etc/passwd` contains information about the accounts on the system. Having audit records for access to account-related files provides evidence of who accessed them and can also help identify suspicious or unauthorized activity.

One issue I noticed is that the audit key is called `passwd_changes`, but the rule also logs read-only access. This could create a lot of routine noise, so the rule may need to be tuned to focus more on actual changes.

Module 2 Log Management and Analysis

Activity 2.1 Explore Logs with journalctl



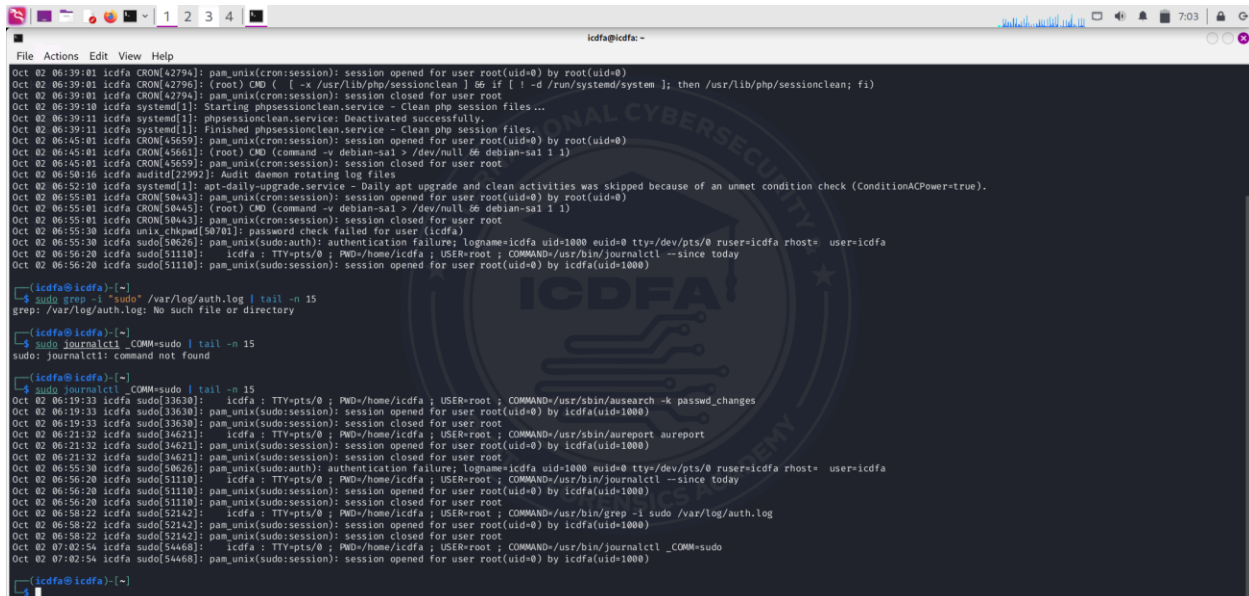
```
icdfa@icdfa:~$ ls /var/log/auth.log /var/log/syslog
ls: cannot access '/var/log/auth.log': No such file or directory
ls: cannot access '/var/log/syslog': No such file or directory

icdfa@icdfa:~$ journalctl -since "today" | tail -n 25
Oct 02 06:32:47 icdfa systemd[1257]: Starting xfconfd.service - Xfce configuration service...
Oct 02 06:32:47 icdfa dbus-daemon[1287]: [session uid=1000 pid=1287 pidfd=5] Successfully activated service 'org.xfce.Xfconf'
Oct 02 06:32:47 icdfa systemd[1257]: Started xfconfd.service - Xfce configuration service.
Oct 02 06:34:31 icdfa auditd[22992]: Audit daemon rotating log files
Oct 02 06:35:01 icdfa CRON[40901]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Oct 02 06:35:01 icdfa CRON[40908]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1 1)
Oct 02 06:35:02 icdfa CRON[40901]: pam_unix(cron:session): session closed for user root
Oct 02 06:39:01 icdfa CRON[42794]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Oct 02 06:39:01 icdfa CRON[42796]: (root) CMD ( [ -x /usr/lib/php/sessionclean ] && if [ ! -d /run/systemd/system ]; then /usr/lib/php/sessionclean; fi)
Oct 02 06:39:01 icdfa CRON[42794]: pam_unix(cron:session): session closed for user root
Oct 02 06:39:10 icdfa systemd[1]: Starting phpsessionclean.service - Clean php session files...
Oct 02 06:39:11 icdfa systemd[1]: phpsessionclean.service: Deactivated successfully.
Oct 02 06:39:11 icdfa systemd[1]: Finished phpsessionclean.service - Clean php session files.
Oct 02 06:45:01 icdfa CRON[45659]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Oct 02 06:45:01 icdfa CRON[45661]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1 1)
Oct 02 06:45:01 icdfa CRON[45659]: pam_unix(cron:session): session closed for user root
Oct 02 06:50:16 icdfa auditd[22992]: Audit daemon rotating log files
Oct 02 06:52:10 icdfa systemd[1]: apt-daily-upgrade.service - Daily apt upgrade and clean activities was skipped because of an unmet condition check (ConditionACPower=true).
Oct 02 06:55:01 icdfa CRON[50443]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Oct 02 06:55:01 icdfa CRON[50445]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1 1)
Oct 02 06:55:01 icdfa CRON[50443]: pam_unix(cron:session): session closed for user root
Oct 02 06:55:30 icdfa unix_chkpmd[50701]: password check failed for user (icdfa)
Oct 02 06:55:30 icdfa sudo[50626]: pam_unix(sudo:auth): authentication failure; logname=icdfa uid=1000 euid=0 tty=/dev/pts/0 ruser=icdfa rhost= user=icdfa
Oct 02 06:56:20 icdfa sudo[51110]: icdfa : TTY=pts/0 ; PWD=/home/icdfa ; USER=root ; COMMAND=/usr/bin/journalctl --since today
Oct 02 06:56:20 icdfa sudo[51110]: pam_unix(sudo:session): session opened for user root(uid=0) by icdfa(uid=1000)

icdfa@icdfa:~$
```

Figure 2-1_journalctl

Activity 2.2 Authentication and Privilege-Use Analysis



```
icdfa@icdfa:~$ journalctl -COMM=sudo | tail -n 15
grep: /var/log/auth.log: No such file or directory

icdfa@icdfa:~$ sudo journalctl -COMM=sudo | tail -n 15
sudo: journalctl: command not found

icdfa@icdfa:~$ journalctl -COMM=sudo | tail -n 15
Oct 02 06:19:33 icdfa sudo[33630]: icdfa : TTY=pts/0 ; PWD=/home/icdfa ; USER=root ; COMMAND=/usr/sbin/ausearch -k passwd_changes
Oct 02 06:19:33 icdfa sudo[33630]: pam_unix(sudo:session): session opened for user root(uid=0) by icdfa(uid=1000)
Oct 02 06:19:33 icdfa sudo[33630]: pam_unix(sudo:session): session closed for user root
Oct 02 06:21:32 icdfa sudo[34621]: icdfa : TTY=pts/0 ; PWD=/home/icdfa ; USER=root ; COMMAND=/usr/sbin/ausearch aureport
Oct 02 06:21:32 icdfa sudo[34621]: pam_unix(sudo:session): session opened for user root(uid=0) by icdfa(uid=1000)
Oct 02 06:21:32 icdfa sudo[34621]: pam_unix(sudo:session): session closed for user root
Oct 02 06:55:30 icdfa sudo[50626]: pam_unix(sudo:auth): authentication failure; logname=icdfa uid=1000 euid=0 tty=/dev/pts/0 ruser=icdfa rhost= user=icdfa
Oct 02 06:56:20 icdfa sudo[51110]: icdfa : TTY=pts/0 ; PWD=/home/icdfa ; USER=root ; COMMAND=/usr/bin/journalctl --since today
Oct 02 06:56:20 icdfa sudo[51110]: pam_unix(sudo:session): session opened for user root(uid=0) by icdfa(uid=1000)
Oct 02 06:58:22 icdfa sudo[52142]: icdfa : TTY=pts/0 ; PWD=/home/icdfa ; USER=root ; COMMAND=/usr/bin/grep -l sudo /var/log/auth.log
Oct 02 06:58:22 icdfa sudo[52142]: pam_unix(sudo:session): session opened for user root(uid=0) by icdfa(uid=1000)
Oct 02 07:02:54 icdfa sudo[54468]: icdfa : TTY=pts/0 ; PWD=/home/icdfa ; USER=root ; COMMAND=/usr/bin/journalctl -COMM=sudo
Oct 02 07:02:54 icdfa sudo[54468]: pam_unix(sudo:session): session opened for user root(uid=0) by icdfa(uid=1000)

icdfa@icdfa:~$
```

Figure 2-2_auth_privilege

Activity 2.3 General System Log Review

```

lcdfa@icdfa:~$ sudo grep -i "error" /var/log/syslog | tail -n 15
grep: /var/log/syslog: No such file or directory

lcdfa@icdfa:~$ sudo journalctl -p err | tail -n 15
Aug 09 18:29:31 icdfa obexd[1814]: Unable to acquire registry: Error calling StartServiceByName for org.gnome.evolution.dataserver.Sources5: Unit evolution-source-registry.service not found.
Aug 09 18:29:31 icdfa obexd[1814]: Unable to acquire registry: Error calling StartServiceByName for org.gnome.evolution.dataserver.Sources5: Unit evolution-source-registry.service not found.
-- Root f33284c2f6134837a3380b424ceda9b0 --
Oct 02 05:11:29 icdfa kernel: RETbleed: WARNING: Spectre v2 mitigation leaves CPU vulnerable to RETbleed attacks, data leaks possible!
Oct 02 05:11:29 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running on an unsupported hypervisor.
Oct 02 05:11:29 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is likely broken.
Oct 02 05:11:29 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please switch to a supported graphics device to avoid problems.
Oct 02 05:11:48 icdfa lightdm[1244]: gkr-pam: unable to locate daemon control file
Oct 02 05:11:48 icdfa lightdm[1008]: pam_systemd(lightdm-greeter:session): failed to release session: Transport endpoint is not connected
Oct 02 05:12:01 icdfa obexd[1795]: Unable to acquire registry: Error calling StartServiceByName for org.gnome.evolution.dataserver.Sources5: Unit evolution-source-registry.service not found.
Oct 02 05:12:01 icdfa obexd[1795]: Unable to acquire registry: Error calling StartServiceByName for org.gnome.evolution.dataserver.Sources5: Unit evolution-source-registry.service not found.
Oct 02 06:05:14 icdfa kernel: watchdog: BUG: soft lockup - CPU#0 stuck for 235s! [swapper/0:0]
Oct 02 06:05:14 icdfa systemd[1]: systemd-udevd.service: Watchdog timeout (limit 3min)!
Oct 02 06:05:14 icdfa systemd[1]: systemd-logind.service: Watchdog timeout (limit 3min)!
Oct 02 06:05:14 icdfa systemd[1]: systemd-journald.service: Watchdog timeout (limit 3min)!

lcdfa@icdfa:~$

```

Figure 2-3_errors_warnings

A short table containing Event/Condition, Evidence Source, Time, User/Service, Security Significance and Recommended Action

Event	Source	Time	User/Service	Significance	Action
Failed sudo password, then access	journalctl	06:55:30 (success 06:56:20)	lcdfa / sudo	One failed privilege login, likely a typo. Repeats could mean password guessing	Alert on 5+ failure in 10 mins
Sudo commands run as root	Journalctl _COMM=sudo	06:19:33 to 07:02:54	lcdfa as root	Every root action is traced to a name user	Review sudo logs regularly
Auth.log and syslog missing	/ls output	2 Oct, ~06:55	System	Evidence only exists in the journal. My auth-failures audit rule watches a file that isn't there	Confirm journal is persistent; forward logs centrally
Kernel soft lockup, watchdog timeouts	Journalctl -p err	06:05:14	Kernel, journald, logind	System stuck ~235s. A journald timeout may mean a gap in logs	Check VM resources; check the journal for gaps
RETbleed CPU vulnerability warning	Journalctl -p err	05:11:29	Kernel	Mitigation incomplete, so data leaks are possible	Check kernel/microcode updates; track as a vulnerability

Module 3 Linux Security Assessment with Lynis

Activity 3.1 Install or Access Lynis

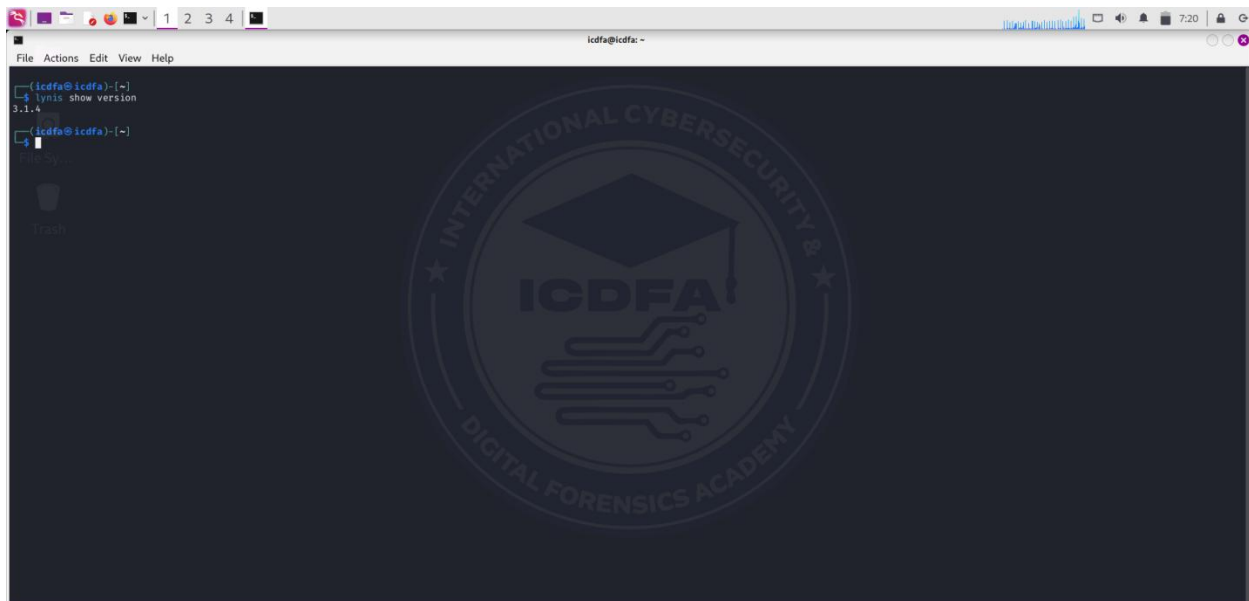


Figure 3-1_lynis_version

Activity 3.2 Run a System Audit

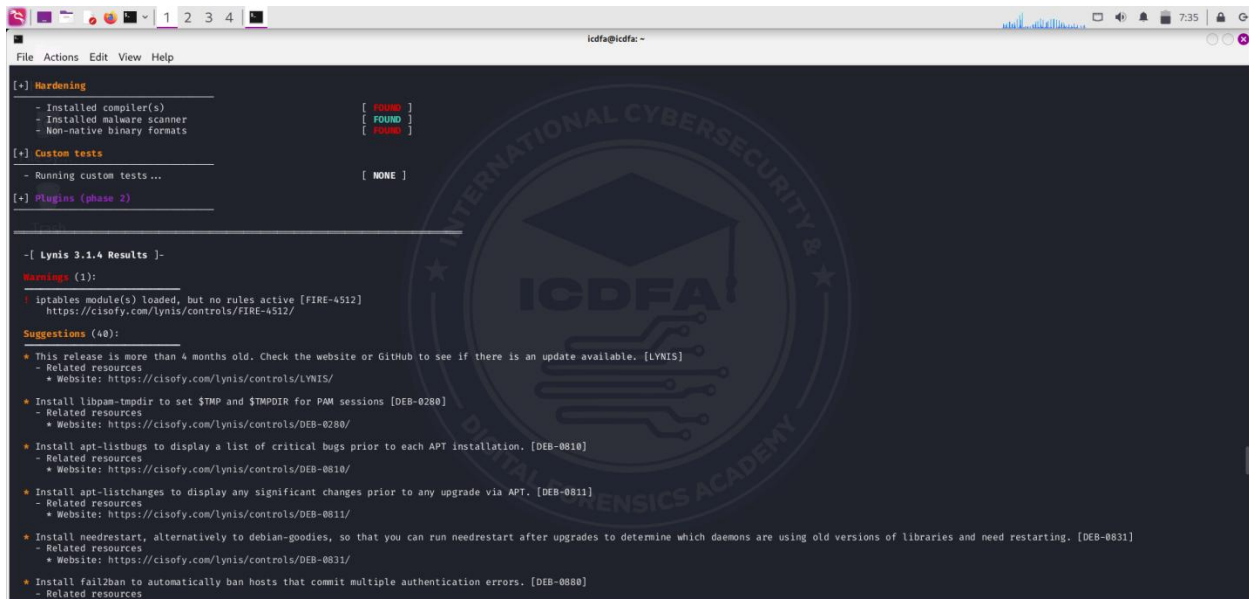


Figure 3-2_lynis-result

```
File Actions Edit View Help

- Show details of a test (lynis show details TEST-ID)
- Check the logfile for all details (less /var/log/lynis.log)
- Read security controls texts (https://cisofy.com)
- Use --upload to upload data to central system (Lynis Enterprise users)

Lynis security scan details:
Hardening index : 65 [#####]
Tests performed : 273
Plugins enabled : 1

Components:
- Firewall [V]
- Malware scanner [V]

Scan mode:
Normal [V] Forensics [ ] Integration [ ] Pentest [ ]

Lynis modules:
- Compliance status [?]
- Security audit [V]
- Vulnerability scan [V]

Files:
- Test and debug information : /var/log/lynis.log
- Report data : /var/log/lynis-report.dat

Lynis 3.1.4
Auditing, system hardening, and compliance for UNIX-based systems
(Linux, macOS, BSD, and others)

2007-2024, CISofy - https://cisofy.com/lynis/
Enterprise support available (compliance, plugins, interface and tools)

[Tip]: Enhance Lynis audits by adding your settings to custom.prf (see /etc/lynis/default.prf for all settings)

(icdfa@icdfa)-[~]
```

Figure 3-2_lynis_summary

```
File Actions Edit View Help

- Malware scanner [V]

Scan mode:
Normal [V] Forensics [ ] Integration [ ] Pentest [ ]

Lynis modules:
- Compliance status [?]
- Security audit [V]
- Vulnerability scan [V]

Files:
- Test and debug information : /var/log/lynis.log
- Report data : /var/log/lynis-report.dat

Lynis 3.1.4
Auditing, system hardening, and compliance for UNIX-based systems
(Linux, macOS, BSD, and others)

2007-2024, CISofy - https://cisofy.com/lynis/
Enterprise support available (compliance, plugins, interface and tools)

[Tip]: Enhance Lynis audits by adding your settings to custom.prf (see /etc/lynis/default.prf for all settings)

(icdfa@icdfa)-[~]
$ sudo grep -E -m 12 --warning suggestion /var/log/lynis-report.dat | head -n 12
[sudo] password for icdfa:
suggestion[]-LVNISI This release is more than 4 months old. Check the website or GitHub to see if there is an update available.|-|-|
suggestion[]-DEB-0280 Install libpam-tmpdir to set TMP and TMPDIR for PAM sessions|-|-|
suggestion[]-DEB-0810 Install apt-listbugs to display a list of critical bugs prior to each APT installation.|-|-|
suggestion[]-DEB-0811 Install apt-listchanges to display any significant changes prior to any upgrade via APT.|-|-|
suggestion[]-DEB-0831 Install needrestart, alternatively to debian-goodies, so that you can run needrestart after upgrades to determine which daemons are using old versions of libraries and need restarting.|-|-|
suggestion[]-DEB-0880 Install fail2ban to automatically ban hosts that commit multiple authentication errors.|-|-|
suggestion[]-ROOT-5122 Set a password on GRUB boot loader to prevent altering boot configuration (e.g. boot in single user mode without password).|-|-|
suggestion[]-ROOT-5180 Determine runlevel and services at startup.|-|-|
suggestion[]-ROOT-5264 Consider hardening system services. Run '/usr/bin/systemd-analyze security SERVICE' for each service.|-|-|
suggestion[]-AUTH-9230 Configure password hashing rounds in /etc/login.defs.|-|-|
suggestion[]-AUTH-9262 Install a PAM module for password strength testing like pam_cracklib or pam_passwdqc or libpam-passwdqc.|-|-|
suggestion[]-AUTH-9282 When possible set expire dates for all password protected accounts.|-|-|

(icdfa@icdfa)-[~]
```

Figure 3-2_lynis_findings

Findings

Findings	Risk / Control Issue	Evidence	Likely owner	Recommended remediation	Priority
FIRE-4512: iptables modules loaded, no rules active	Network filtering is not enforced, so running services could be exposed	Lynis warning in the result section	System / Network Administrator	Define a default-deny rule set, enable it, and check with <i>iptables -L</i>	High
AUTH-9262 and AUTH-9282: no password strength module, no expiry dates on password-protected accounts	Weak or never-expiring passwords weaken access control	Suggestions in the findings session	Identity and Access Management / System Administrator	Install a PAM password-quality module and set expiry rules in <i>/etc/login.defs</i>	Moderate
DEB-0880: fail2ban not installed	No automatic response to repeated failed logins over the network	Suggestion in the findings session	Security Operations	Install and configure fail2ban, and forward its alerts to the SIEM	Moderate
BOOT-5122: no password set on the GRUB boot loader	Anyone with console access could change boot settings, for example boot into single-user mode without a password	Suggestion in the findings session	System Administrator	Set a GRUB password and protect the boot configuration	Moderate

Module 4 SIEM, Automation and Continuous Monitoring

SIEM and Continuous monitoring mapping

Evidence	How it reaches the SIEM	Stays a technical alert	Becomes a governance issue
Auditd:/etc/passwd, accessed via sudo, key(password changes)	Forward audit logs with a log agent, parsed by the key and user(auid)	A single read by a known user	Repeated or unexpected writes to account files with no approved change; no audit coverage on critical files
Journalctl: failed sudo password at 06:55:30, then success	Forward journal entries; correlate by the user and time	One failed attempt that is followed by a success	Repeated failures across accounts or hosts, or privileged use outside approved times with no owner
Missing: auth.log and syslog(journal only)	Confirm the journal is persistent and forwarded centrally	A one-off collection error	Logging that isn't centrally retained, which makes evidence unreliable
Lynis: hardening index 65, warning FIRE-4512,40 suggestions	Scheduled scan, with the report data ingested and tracked over time	A single suggestion	An unresolved High finding(no active firewall rules) past its remediation date, or a falling score

How automation helps

1. Automated alerting: the SIEM alerts when a rule matches, such as failed sudo attempts above a threshold.
2. Continuous control monitoring: scheduled Lynis scans and audit checks confirm the controls still work.
3. Workflow automation: a finding is assigned to an owner, tracked, and escalated if it isn't fixed in time.
4. Trend analysis and reporting: the hardening index and failed-login counts are tracked and reported to management.

Governance discussion

Operational alerts cover individual events, such as one failed sudo attempt or one read of */etc/passwd*. These go to the SOC to triage. A condition becomes a governance or control-assurance issue when it shows a control that is missing, failing or not owned. Examples are no active firewall rules, logs that aren't centrally retained, or a High finding left open past its deadline. These need a named owner, a remediation date and a retest, and they are reported to management.

Applied Governance Task from Linux Evidence to Control Assurance

Governance issues identified

1. Host firewall not enforced (High): The Lynis scan raised warning FIRE-4512: iptables modules are loaded, but no rules are active. This was the only warning in the scan. Without active rules, network traffic is not filtered, so any listening service could be exposed.
2. Incomplete log coverage (Moderate): */var/log/auth.log* and */var/log/syslog* do not exist on this system, so I used the systemd journal instead. The journal worked, but it is now the only source of authentication and system evidence. If it is not kept and forwarded centrally, audit evidence could be lost.
3. Weak preventive hardening (Moderate): Lynis scored the system 65, with 273 tests, 1 warning and 40 suggestions findings include no GRUB password (BOOT-5122), no password-strength module (AUTH-9262), no password expiry dates (AUTH-9282) and no fail2ban (DEB-0880)). None of these show an attack, but together they weaken access control.

I also observed one failed sudo password at 06:55:30, followed by a successful one at 06:56:20. One failure is not a governance issue. It becomes one only if it repeats.

Control-Monitoring Table

Control / Objective	Evidence	Owner	Observed Status
Monitor access to <i>/etc/passwd</i>	1-3_ausearch_passwd: access at 6:06:42 by icdfa via sudo, read only	System Admin	Operating; rule loaded, events captured
Detect failed privileged logins	2-1_journalctl: failed sudo at 6:55:30, success at 6:56:20	Security Operation	Detected; one failure, no repeats
Auth and system logs available and retain	2-1_Journalctl: auth.log and syslog not found; journal used	System Admin	Gap; journal is the only source
Host firewall enforced	3-2_lynis_result: FIRE-4512, iptables loaded, no rules active	System/ Network Admin	Not effective
System meets hardening baseline	3-2_lynis_summary and 3-2_findings (BOOT-5122, AUTH-9262, AUTH-9282, DEB-0880)	System Admin/IAM	Partially hardened

Governance Escalation Questions

1. Which finding is the highest priority and why?

FIRE-4512 is the highest priority. It was the only warning in the Lynis scan, and there are no active firewall rules, which means network traffic is not being filtered.

The other findings, such as the missing GRUB password and password expiry issues, are still weaknesses that need to be fixed, but they generally require console access or user action. Because of this, I would treat the firewall issue as more urgent.

2. Who should own remediation for each major finding?

- Firewall (FIRE-4512): System / Network Administrator
- Password strength and expiry (AUTH-9262, AUTH-9282): Identity and Access Management / System Administrator
- GRUB password (BOOT-5122): System Administrator
- Log retention and forwarding: System Administrator / Logging Team
- fail2ban and failed-login alerting: Security Operations

The CISO or Risk function should be accountable for tracking the findings and making sure they are properly closed.

3. What should trigger escalation to the CISO, Risk function or management committee?

Escalation should happen when there is a serious or repeated security issue, for example:

- A High finding remains open after its agreed deadline, such as 7 days.
- There are repeated failed privileged logins, such as 5 or more within 10 minutes, or privileged activity happens outside approved times.
- There are unapproved writes to account files, such as /etc/passwd.
- Audit logging is missing or stops working on a critical system.
- The hardening index drops during a later scan.

These thresholds are proposed for this assessment and would need to be formally agreed and documented in the organisation's policy.

4. What evidence would show that the remediation worked?

For each finding, I would look for evidence that the issue has actually been fixed:

- Firewall: iptables -L showing active firewall rules, followed by a new Lynis scan showing that FIRE-4512 is no longer reported.
- Passwords: Evidence of the new PAM module and password expiry settings in the configuration, with the AUTH findings no longer appearing in Lynis.
- GRUB: A new Lynis scan showing that BOOT-5122 no longer appears.

- Logging: Evidence that the journal keeps older entries after a reboot and that logs are successfully being forwarded to the central store.
- Overall: A hardening index higher than the 65 recorded in this lab.

5. When and how should the control be retested?

I would retest each control immediately after the fix is applied by rerunning Lynis and comparing the results with the original baseline.

After the initial retest, I would run scheduled Lynis scans monthly and regularly review the audit and journal logs to make sure the controls are still working.

High-priority findings should be retested within a few days after remediation. The results should be recorded in the control-monitoring table, and if a finding appears again, it should be returned to the responsible owner for action and escalated where necessary.